



Haga Evidence Packet

The Hook

Haga ran your exact task under an independent physics stress benchmark. Reproducible. Seeded. Failure cases shown below:

Metric	Baseline (Normal)	Severe Stress	Delta
Success Rate			
95% CI (Wilson)			
Seeds × Condition	50 × 1	50 × 4	200 Evaluations
Failure Mode			Video Attached

Stress conditions tested:

- **Mass perturbation:** $\pm 50\%$ object mass
- **Frictional sweep:** $0.5x \rightarrow 2.0x$ nominal
- **Actuator delay:** $0 \rightarrow 100\text{ms}$ injection
- **Disturbance forces:** Random impulses $0.5 - 2.0\text{N}$

Why this matters for you

Sim benchmarks report 89.4% success; real robots achieve 12% on household tasks (Stanford Index 2026). That gap is unverified physics.

Your policy looks robust in normal simulation. **Physics stress reveals the failure mode that kills sim-to-real transfer** - before you deploy.

What you get as a Design Partner (\$5k/month)

Deliverable	Frequency	Format
Physics Stress Report	Weekly	PDF + JSON + failure videos
CI Integration	One-time setup	Github Actions / Gitlab CI / Jenkins
Regression Alerting	Per commit	Slack/Email on Physics Regression

Methodology Access	Ongoing	Full detector definitions, thresholds
Priority Support	Ongoing	Direct Slack/Email, 24h SLA

Pilot: 1 month, \$5k. Cancel anytime. No annual contract.

Your Specific Results (Pre-computed for this outreach)

Task:

Policy:

Environment:

Seeds:

Conditions:

HEADLINE RESULT:

Nominal Success:

Severe Stress:

Absolute Drop:

PER-CONDITION BREAKDOWN:

Condition	Success	95% CI	n
Nominal			50
Mass +50%			50
Mass -50%			50
Friction 0.5x			50
Friction 2.0x			50
Actuator Delay (100ms)			50

Methodology & Credibility

Haga’s adversarial physics stress methodology. One methodology. Two artifacts. Same evidence standard.

Pillar 1: Policy Stress Testing

INPUT: Robot policy (checkpoint, configuration, task specifications)

STEP 1: Artifact Verification

- Parse policy + environment + task definition
- Lock reproducible run plan (seeds, versions, hashes)
- Acceptance criteria: 1.000 deterministic replay

STEP 2: Adversarial Condition Generation

- Physics perturbations (mass, friction, delay, disturbances, sensors, actuator saturation)
- Tiered severity: Nominal → moderate → severe → extreme
- Paired seeds: Same seed across all conditions

STEP 3: Calibrated Execution

- 50+ seeds per condition (Wilson 95% CI)
- Fixed compute budget per evaluation
- Failure capture: Video + state trace + detector log

STEP 4: Evidence-Graded Reporting

- Numeric: Success rate, CI, effect size, p-value
- Qualitative: Failure mode taxonomy + video clips
- Actionable: "Fix X → expect Y% recovery"

Pillar 2: World-Model Physics Consistency (Bonus)

Same detectors, applied to generated video. Your generative world models (if any) can be checked with the same infrastructure.

- **Teleportation detection:** 1.000 recall, 0 FP (calibrated)
- **Anti-gravity detector:** Catches floating objects
- **Causeless impulse detector:** Spontaneous velocity changes
- **Interpenetration detector:** Geometry violations

Live Evidence (Public, Reproducible)

Benchmark	Task	Nominal → Severe	Detector
Robosuite Lift	SAC Policy	1.00 → 0.26	Mass/Friction/Delay
Robosuite Stack	SAC Policy	0.96 → 0.20	Mass/Friction/Delay
Robosuite PickPlaceCan	SAC Policy	1.00 → 0.24	Mass/Friction/Delay
Robosuite Door	SAC Policy	0.60 → 0.52	Mass/Friction/Delay

CogVideoX 12V	Static hover	0% → 100% fall	Physics-IQ (static_hover)
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Why Independent Verification

Self-Reported	Haga Independent
Cherry-picked seeds	Fixed seeds (0 - 49), published
Nominal conditions only	Adversarial physics sweep
"Success Rate: 94%"	"94% [91-96%] nominal → 20% [15-26%] severe"
No failure videos	Every failure: video + static trace
Methodology hidden	Full detector definitions + thresholds public
Conflict of interest	No stake in your policy performance

Pilot Terms & Next Steps

Design Partner Pilot: \$5k for 1 Month

SCOPE

- 4 tasks of your choice (Lift, Stack, PickPlace, etc.)
- Weekly stress reports (PDF + JSON + videos)
- CI integration (GitHub Actions / GitLab / Jenkins)
- Regression alerting on every commit

DELIVERABLES (Week 1 - 4)

- Week 1: Baseline + severe stress on Task 1
- Week 2: Task 2 + CI Integration
- Week 3: Task 3 + regression alerting live
- Week 4: Task 4 + methodology handoff + renewal talk

PAYMENT

- \$5,000 USD upfront (Wise/Bank transfer)
- No auto-renewal: Explicit opt-in for Month 2
- Cancel anytime, prorated refund for undelivered weeks

YOUR COMMITMENT

- Policy checkpoints + configurations for 4 tasks
- Environment specifications (MuJoCo version, assets, seeds)

- 30-min kickoff call + 15-min weekly sync
- Feedback on report format / actionability

Next Steps (If Interested)

1. **Reply "Yes"** → I send Google Meet for 30-min kickoff (this week)
2. **Kickoff call** → Confirm 4 tasks, share checkpoints, lock seeds
3. **Day 1 - 3** → I run Task 1 baseline + stress + send first report
4. **Weekly** → New task report + CI Integration process
5. **Month end** → Renew / expand / part ways - no pressure

Contact

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- Slack Connect available for **design partners**

"We don't replace your sim. We stress it until it breaks — so you know exactly where your policy fails before the robot does."